Experiment: Domain adaption of GPT-2 model

1. Objective

Investigate how well continued pretraining of a GPT-2 model on theoretical physics papers improves its ability to predict domain-specific text. Training progress is monitored by calculating the cross-entropy loss for training and validation data during pretraining.

2. The model

The experiment uses GPT-2 Small (124 million parameters), initialized with pretrained OpenAI weights. The model and pretraining settings are summarized below.

GPT-2 model settings

Vocabulary size	50 257
Context length	1024
Embedding dimension	768
# of attention heads	12
# of layers	12
Dropout rate	0.1

Pretraining settings

Number of epochs	2
Batch size	4
Evaluation frequency	50
Evaluation batches	20
Learning rate	1e-4
Weight decay	0.1

3. The dataset

The corpus consists of 113 theoretical physics papers from the research areas “Swampland” and “Quantum Gravity”, all published on arXiv.org within the last three years. Five papers are reserved for validation, while the remaining 108 are used for training. Over two training epochs, the model processes approximately seven million tokens. The papers are downloaded as LaTeX source files, which are extracted and preprocessed to remove common formatting commands while keeping the scientific text.

4. Results and interpretation

Both the training and validation loss decrease over the course of pretraining, with training loss dropping from 2.904 to 1.899 and validation loss from 2.883 to 2.045 (see figure 1 on the next page).

Training loss exhibits stronger fluctuations, possibly reflecting that LaTeX files are heterogeneous datasets. Despite preprocessing, the files contain a mixture of ordinary text, mathematical expressions and residual formatting commands. Validation loss decreases more smoothly and reaches a plateau towards the end of training. The results indicate improved performance in next-token prediction, but visible coherence of generated text likely requires a much larger training corpus.

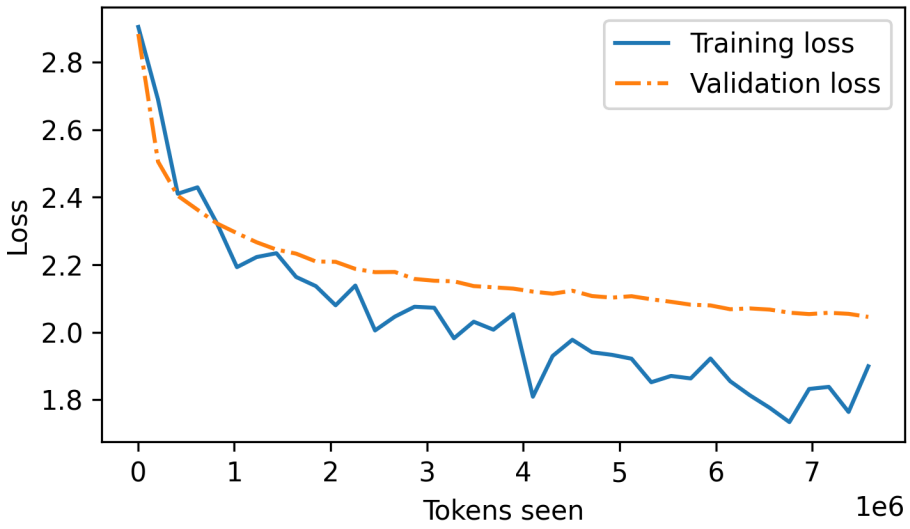


Figure 1: Training and validation loss plotted against the number of tokens seen during pretraining.